

Fuel-Optimal Boost-back Guidance Algorithms for Reusable Launch Vehicles

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Abstract

This paper presents two new guidance algorithms for the boost-back phase of reusable launch vehicles (RLVs): 1) the instantaneous impact point (IIP) guidance considering flight path angle and 2) the minimum-impulse guidance. The two algorithms aiming to minimize propellant consumption are applicable to both downrange landing (DRL) and return-to-launch-site (RTLS) scenarios. The first modifies a previously developed IIP guidance algorithm by incorporating the flight path angle control, enhancing its adaptability and performance while preserving the analytic integrity of the original formulations. The second algorithm utilizes the hit equation-based formulations to determine the minimum velocity increments through two distinct approaches. Case studies on the boost-back phase of an RLV in DRL and RTLS scenarios validate the effectiveness of the proposed algorithms. Furthermore, a comprehensive analysis highlights the trade-offs between computational cost and performance, offering practical insights into selecting a proper guidance algorithm for given mission requirements.

Keywords: Reusable Launch Vehicle (RLV), Boost-Back Burn, Return To Launch Site (RTLS), Downrange Landing (DRL), Lambert Problem, Instantaneous Impact Point (IIP)

Nomenclature

$\mathbf{a}_T, a_{T,1}, a_{T,2}, a_{T,3}$	Acceleration vector generated by thrust and its components
C_I^E	Transform matrix from the ECI frame to the ECEF frame
$\mathbf{d}_v, \mathbf{d}_{\dot{\gamma}}, \mathbf{d}_{\dot{\chi}}$	The sensitivity vectors of the velocity component rates on the IIP rate
Δv	Velocity increment (del-v)
$\mathbf{g}$	Gravity acceleration vector
$\mathbf{i}_h$	Unit vector of angular momentum vector
$\mathbf{i}_u$	Unit vector for shortest path

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$\mathbf{i}_q$	Unit vector normal to the plane defined by $\mathbf{i}_p$ and $\mathbf{i}_l$
Lat, Lon	Latitude and Longitude
$m, \dot{m}$	Mass and mass flow rate
$\mathbf{r}, r$	Position vector and its magnitude
r_p	Radius of the Earth
$\mathbf{r}_p, \mathbf{i}_p, d\mathbf{i}_p/dt$	IIP vector, its unit vector, and time derivative of $\mathbf{i}_p$
t, t_p, t_{ref}	Current time, time of flight, and reference time
T	Thrust magnitude
$\mathbf{V}, v$	Velocity vector and its magnitude
v_{ex}	Effective exhaust velocity
$\chi, \dot{\chi}$	Azimuth angle and its rate
$\gamma, \dot{\gamma}$	Flight path angle and its rate
ω_E	Angular velocity of the Earth
ϕ	Flight angle
μ	Standard gravitational parameter of the Earth

Subscripts

0	Value for the rocket's current state
d	Desired value
g	Value affected by gravity
t	Target value
T	Value affected by thrust

Superscripts

*	Optimal value
E	Value represented in the ECEF frame

1. Introduction

Reusable Launch Vehicles (RLVs) have revolutionized modern space transportation by dramatically reducing launch costs and increasing mission frequency. The descent sequence of a reusable launch vehicle (RLV) involves three burns: the boost-back burn, the re-entry burn, and the terminal landing burn [1-3]. As the total impulse for the boost-back burn is dominant among the three burns, propellant consumption during this phase significantly influences payload capacity and mission feasibility – particularly for the return to launch site (RTLS) scenario. The primary goal of the boost-back burn is to re-orient a reusable stage towards the designated landing site, thereby effectively shifting the instantaneous impact point (IIP) to the vicinity of the landing site. Moreover, it aims to reduce the horizontal velocity, which is especially important for the downrange landing, to simplify the subsequent maneuvers for a vertical landing [4]. Fig. 1 shows the conceptual diagram of the boost-back phase of an RLV.

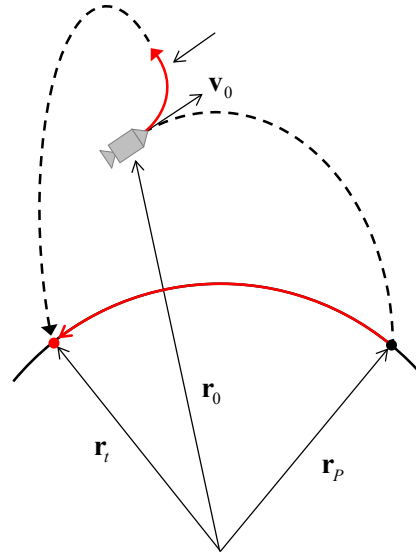


Fig. 1 Conceptual diagram of the boost-back guidance of an RLV

The development of advanced, fuel-optimal guidance algorithms tailored for the boost-back burn phase remains relatively underexplored, particularly for Vertical Take-Off and Vertical Landing (VTVL) configurations. A few past studies explored the algorithms for winged RLVs [5-8], providing insights that partially apply to VTVL RLVs. Jo and Ahn [9] proposed an instantaneous impact point (IIP) guidance algorithm for VTVL RLVs, which utilized the IIP dynamics to deliver a non-iterative, closed-form acceleration command. However, they noted the potential performance degradation when applying the algorithm to return-to-launch-site (RTLS) scenarios [10]. Brendel et al. [11] proposed an optimal guidance algorithm for toss-back concepts of RLVs by applying a shooting method based on Pontryagin's Maximum Principle. Their toss-back guidance algorithm is promising but depends on the errors of the current state from the reference trajectory. The boost-back guidance applied to the CALLISTO project computes the attitude command based on a numerical prediction of the drag landing point (DLP), the IIP considering the drag effect [12]. Wei et al. [13] presented an arbitrary-yaw iterative guidance algorithm for the boost-back of VTVL RLVs by utilizing geometry and time updaters, which compensate for the guidance errors. With an acceptable performance under the initial deviations and model uncertainty, their guidance law can cover

the flight requiring a substantial yaw angle maneuver.

This paper proposes two new guidance algorithms for the boost-back phase of reusable launch vehicles (RLVs): 1) the instantaneous impact point (IIP) guidance considering flight path angle and 2) the minimum-impulse guidance. The two algorithms aiming to minimize propellant consumption are applicable to both downrange landing (DRL) and return-to-launch-site (RTLS) scenarios. The first modifies a previously developed IIP guidance algorithm developed by Jo and Ahn [9]. A major limitation of the previous IIP guidance algorithm was its significant performance degradation when applied to the RTLS scenario. The new algorithm incorporated flight path angle (FPA) control into the IIP guidance, enhancing its adaptability and performance while preserving the analytic integrity of the original formulations. The second method proposed in this paper is the minimum-impulse guidance algorithm based on the hit equation [14]. This algorithm calculates the optimal velocity from the two distinct approaches depending on the iterator variable: 1) the FPA and 2) the time of flight (TOF). The first approach constructs an analytic expression for the Δv as a function of FPA. Then, the algorithm finds the FPA that minimizes the Δv by differentiating the expression. The second approach modifies the Lambert guidance [15, 16] by introducing a new steering logic to pursue the desired velocity from the current state efficiently.

The contribution of this study is twofold. First, it proposes guidance algorithms tailored for the boost-back phase of an RLV. The proposed algorithms can provide a closed-form acceleration command with a near-optimal performance in various scenarios. Second, comprehensive case studies using realistic RLV configurations (e.g., Falcon 9) and DRL/RTLS scenarios have been conducted to evaluate the performance of the proposed algorithms. This study can provide a foundation for further enhancing RLV guidance strategies, with implications for future missions that require balancing computational efficiency and trajectory precision.

2. Instantaneous Impact Point Guidance Considering Flight Path Angle

The IIP guidance algorithm changes the IIP of a rocket to the landing site along the shortest path (great circle), showing near-optimal performance [9]. Notably, the performance is almost optimal for the IIP change to the downrange direction. On the other hand, the performance worsens when the change in the reverse downrange direction is required (e.g., RTLS scenario). Jo and Ahn [10] investigated the reason for the performance degradation, and they concluded that the conventional IIP guidance only focuses on the IIP dynamics and always selects the command generating the maximum IIP rate at every guidance cycle regardless of the trajectory shape of the rocket, resulting in performance deterioration in some instances. This section revises the conventional IIP guidance algorithm by adding the flight-path-angle rate control scheme to apply the IIP guidance algorithm to the RTLS scenario without compromising the performance.

2.1. Vehicle Dynamics

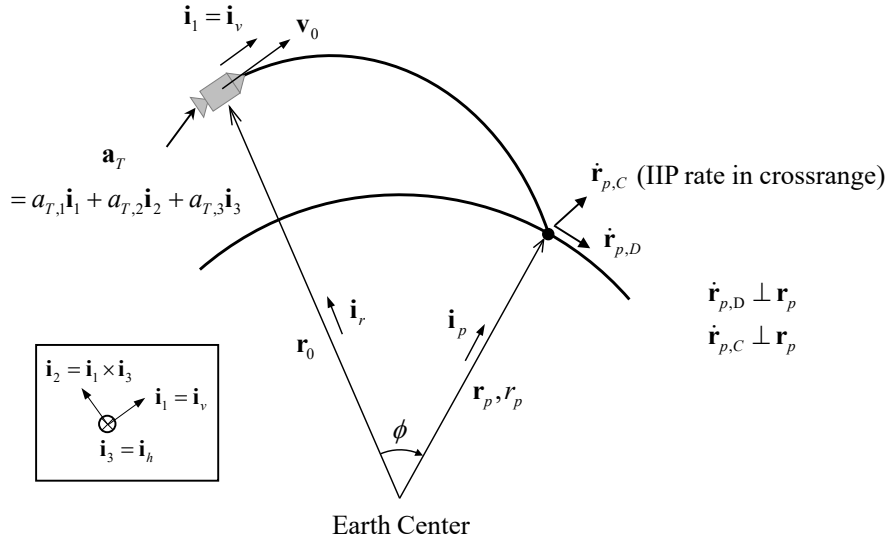


Fig. 2 Geometry for IIP and its rate calculation in the ECI frame

This subsection reviews the IIP dynamics of the rocket described in [17, 18]. Note that quantities with a superscript E are defined in the ECEF frame, whereas those without it are expressed in the ECI frame. The governing equations of the rocket in the Earth Centered Inertial (ECI) frame are expressed as

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}) + \mathbf{a}_T, \quad \dot{m} = -T/v_{ex}. \quad (1)$$

where, $\mathbf{r}$ and $\mathbf{v}$ are the position and velocity vectors of the rocket, respectively. Also, $\mathbf{g}(\mathbf{r})$, m , and v_{ex} are the gravity acceleration vector, mass, and effective exhaust velocity, respectively. $\mathbf{a}_T$ is the acceleration vector generated by thrust (T) and is expressed as

$$\mathbf{a}_T = a_{T,1}\mathbf{i}_1 + a_{T,2}\mathbf{i}_2 + a_{T,3}\mathbf{i}_3. \quad (2)$$

In Eq. (2), $\mathbf{i}_1$ and $\mathbf{i}_3$ are the unit vectors parallel to the velocity and angular momentum vectors, respectively, and $\mathbf{i}_2 (\equiv \mathbf{i}_1 \times \mathbf{i}_3)$ can be calculated from the two unit vectors. Based on our concept of operations (ConOps), the proposed boost-back algorithm moves the Keplerian impact point of the vehicle to the desired location. The error

caused by disturbing forces (e.g., aerodynamics and gravitational perturbations) is corrected during the subsequent phases.

The time derivatives of the velocity components affected by the thrust are defined as

$$\dot{v}_{0,T} = a_{T,1}, \quad \dot{\gamma}_{0,T} = \frac{a_{T,2}}{v_0}, \quad \dot{\chi}_{0,T} = \frac{a_{T,3}}{v_0 \cos \gamma_0}, \quad (3)$$

where v is the magnitude of the velocity vector, γ is the flight path angle (FPA), and χ is the azimuth angle defined in the ECI frame. Note that subscript a denotes the values affected by the thrust acceleration ($\mathbf{a}_T$). The magnitude of the thrust acceleration is expressed as

$$|\mathbf{a}_T| = a_T = \frac{T}{m} = \sqrt{a_{T,1}^2 + a_{T,2}^2 + a_{T,3}^2} = \sqrt{\dot{v}_{0,T}^2 + v_0^2 \dot{\gamma}_{0,T}^2 + (v_0 \cos \gamma_0)^2 \dot{\chi}_{0,T}^2} \quad (4)$$

2.2. Instantaneous Impact Point Dynamics

As shown in Fig. 2, the IIP is an intersection point between the coasting trajectory and the Earth's surface, whose dynamics is expressed as

$$\mathbf{i}_p = \frac{\mathbf{r}_p}{r_p} = \frac{\cos(\phi + \gamma_0)}{\cos \gamma_0} \mathbf{i}_r + \frac{\sin \phi}{\cos \gamma_0} \mathbf{i}_v. \quad (5)$$

$$\begin{aligned} \frac{d}{dt} \mathbf{i}_p &= \left(\frac{\partial \phi}{\partial v_0} \cdot \mathbf{i}_D \right) \dot{v}_{0,T} + \left(\frac{\partial \phi}{\partial \gamma_0} \cdot \mathbf{i}_D \right) \dot{\gamma}_{0,T} + (\sin \phi \cdot \mathbf{i}_C) \dot{\chi}_{0,T} \\ &= \mathbf{d}_v \dot{v}_{0,T} + \mathbf{d}_\gamma \dot{\gamma}_{0,T} + \mathbf{d}_\chi \dot{\chi}_{0,T}, \end{aligned} \quad (6)$$

In Eq. (5), $\mathbf{i}_p$, $\mathbf{i}_r$, and $\mathbf{i}_v$ are the unit vectors of the IIP, position, and velocity vectors of the rocket, respectively. Furthermore, $\mathbf{r}_p$ and r_p are the IIP vector and its magnitude. Assuming that the Earth is a perfect sphere, r_p is the radius of the Earth. The subscript 0 indicates the values corresponding to the rocket's current state. ϕ is the flight angle. In Eq. (6), $\mathbf{d}_v$, $\mathbf{d}_\gamma$ and $\mathbf{d}_\chi$ are the sensitivity vectors representing the influence of the rate of velocity components on the IIP rate vector. The flight angle in Eq. (5) is represented as

$$\phi = \sin^{-1} \left(\frac{k_1 k_3 + \sqrt{k_1^2 k_3^2 - (k_1^2 + k_2^2)(k_3^2 - k_2^2)}}{(k_1^2 + k_2^2)} \right), \quad (7)$$

with

$$k_1 = -\frac{h}{\mu r_0} (\mathbf{r}_0 \cdot \mathbf{v}_0), \quad k_2 = \frac{h^2}{\mu r_0} - 1, \quad k_3 = \frac{h^2}{\mu r_p} - 1, \quad (8)$$

where μ and $h (\equiv r_0 v_0 \cos \gamma_0)$ are the standard gravitational parameter of the Earth and angular momentum of the rocket, respectively. The time of flight (TOF), the freefall time of the rocket from the current position to the IIP, can be obtained using the following equation.

$$t_p = \frac{r_0}{v_0 \cos \gamma_0} \left(\frac{\tan \gamma_0 (1 - \cos \phi) + (1 - \Lambda) \sin \phi}{(2 - \Lambda) \left(\frac{1 - \cos \phi}{\Lambda \cos^2 \gamma_0} + \frac{\cos(\gamma_0 + \phi)}{\cos \gamma_0} \right)} + \frac{2 \cos \gamma_0}{\Lambda (2 / \Lambda - 1)^{1.5}} \arctan \left(\frac{\sqrt{2 / \Lambda - 1}}{\cos \gamma_0 \cot(\phi / 2) - \sin \gamma_0} \right) \right), \quad (9)$$

where $\Lambda \equiv (v_0 / v_c)^2 = r_0 v_0^2 / \mu$.

The goal of the guidance scheme is to target the Earth-fixed location (e.g., landing site), the Earth's rotation should be considered in the IIP dynamics as

$$\frac{d}{dt} \mathbf{i}_p^E = \mathbf{d}_{\dot{v}}^E \dot{v}_{0,T} + \mathbf{d}_{\dot{\gamma}}^E \dot{\gamma}_{0,T} + \mathbf{d}_{\dot{\chi}}^E \dot{\chi}_{0,T}, \quad (10)$$

In Eq. (10), $\mathbf{d}_{\dot{v}}^E$, $\mathbf{d}_{\dot{\gamma}}^E$ and $\mathbf{d}_{\dot{\chi}}^E$ are the sensitivity vectore representing the influence of the rate of velocity components on the IIP rate vector in the ECEF frame, expressed as

$$\mathbf{d}_{\dot{v}}^E = \omega_E D_{\dot{v}}^{\dot{p}} \begin{bmatrix} i_{py}^E \\ -i_{px}^E \\ 0 \end{bmatrix} + \mathbf{C}_I^E \mathbf{d}_{\dot{v}}, \quad \mathbf{d}_{\dot{\gamma}}^E = \omega_E D_{\dot{\gamma}}^{\dot{p}} \begin{bmatrix} i_{py}^E \\ -i_{px}^E \\ 0 \end{bmatrix} + \mathbf{C}_I^E \mathbf{d}_{\dot{\gamma}}, \quad \mathbf{d}_{\dot{\chi}}^E = \mathbf{C}_I^E \mathbf{d}_{\dot{\chi}}. \quad (11)$$

where, $\mathbf{i}_p^E = [i_{px}^E, i_{py}^E, i_{pz}^E]$ is the IIP unit vector in the ECEF frame, which is regarded as a target for the guidance scheme. More detailed explanations can be found in the Appendix.

2.3. Shortest Path Following Condition

Using the relationship between the thrust acceleration and IIP rate in Eq. (58), we can directly control the IIP by generating the appropriate acceleration corresponding to the desired direction of the IIP rate vector. The most intuitive method to determine the IIP rate vector's direction is to force the IIP to follow the shortest path between the current and desired IIPs. The direction for the shortest path, or the desired unit vector of the IIP rate ($\mathbf{i}_u^E$), is defined as [9]

$$\mathbf{i}_u^E = \frac{d}{dt} \mathbf{i}_p^E \Big/ \left| \frac{d}{dt} \mathbf{i}_p^E \right| = \mathbf{i}_q^E \times \mathbf{i}_p^E, \quad (12)$$

where $\mathbf{i}_q^E$ is a unit vector normal to the plane defined by the current and target IIPs in the ECEF frame. Note that the IIP vector is always perpendicular to its rate vector. Thus the condition expressed in Eq. (12) leads to the relationship between elements of the IIP rate vector and $\mathbf{i}_u^E$ as follows

$$\left(\frac{d}{dt} i_{px}^E \right) \Big/ \left(\frac{d}{dt} i_{pz}^E \right) = \frac{i_{ux}^E}{i_{uz}^E}. \quad (13)$$

Putting Eq. (56) into Eq. (13) yields

$$\dot{\chi}_{0,T} = A \dot{v}_{0,T} + B \dot{\gamma}_{0,T}, \quad (14)$$

where,

$$A = \frac{(i_{ux}^E / i_{uz}^E) d_{\dot{v},z}^E - d_{\dot{v},x}^E}{d_{\dot{\chi},x}^E - (i_{ux}^E / i_{uz}^E) d_{\dot{\chi},z}^E}, \quad B = \frac{(i_{ux}^E / i_{uz}^E) d_{\dot{\gamma},z}^E - d_{\dot{\gamma},x}^E}{d_{\dot{\chi},x}^E - (i_{ux}^E / i_{uz}^E) d_{\dot{\chi},z}^E}. \quad (15)$$

Combining Eqs. (4) and (14) yields

$$(1 + A^2 v_0^2 \cos^2 \gamma_0) \dot{v}_{0,T}^2 + (v_0^2 + B^2 v_0^2 \cos^2 \gamma_0) \dot{\gamma}_{0,T}^2 + (2AB v_0^2 \cos^2 \gamma_0) \dot{v}_{0,T} \dot{\gamma}_{0,T} = a_T^2 \quad (16)$$

Eq. (16) is the condition to enforce the IIP to follow the shortest path connecting the current IIP and the target

location.

2.4. Flight Path Angle Rate Control

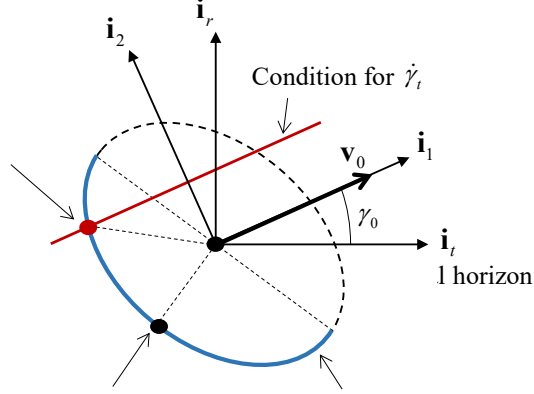


Fig. 3 Concept of IIP guidance with the flight-path-angle rate control

Fig. 3 illustrates the admissible set of guidance commands that change the current IIP to the target location along the shortest path. Commands below the i_t axis decrease the FPA as indicated in Eq. (3), which leads to performance degradation in the RTLS scenario. If the command associated with the maximum IIP rate falls within this region, the IIP guidance algorithm generates a descending trajectory. To address this issue, we propose an algorithm that generates the acceleration command to achieve a specified FPA rather than focusing solely on maximizing the IIP rate. Since the shortest path condition enforces the IIP to move toward the target location, the proposed algorithm selects the point that simultaneously satisfies two conditions for the shortest path and desired FPA rate, as shown in Fig. 3, thereby controlling the rocket's trajectory and the IIP effectively. The flight-path-angle rate produced by the acceleration is expressed as

$$\dot{\gamma} = \dot{\gamma}_{0,g} + \dot{\gamma}_{0,T} = \frac{\left(\frac{v_0^2}{r_0} - \frac{\mu}{r_0^2} \right) \cos \gamma_0}{v_0} + \dot{\gamma}_{0,T}, \quad (17)$$

where $\dot{\gamma}_{0,g}$ is the flight-path-angle rate affected by the gravitational acceleration. To achieve the target flight-path-angle rate, $\dot{\gamma}_{0,a}$ should satisfy the following equation obtained by rearranging Eq. (17)

$$\dot{\gamma}_{0,T} = \dot{\gamma}_t - \frac{\left(\frac{v_0^2}{r_0} - \frac{\mu}{r_0^2} \right) \cos \gamma_0}{v_0}. \quad (18)$$

Since $\dot{\gamma}_{0,a}$ is already determined from Eq. (18) with the given desired FPA rate ($\dot{\gamma}_t$), an analytic expression for $\dot{v}_{0,a}$ can be obtained from Eq. (16) as

$$\dot{v}_{0,T} = \frac{-ABv_0^2 \cos^2 \gamma_0 \dot{\gamma}_{0,T} \pm \sqrt{-(1+A^2v_0^2 \cos^2 \gamma_0 + B^2 \cos^2 \gamma_0)v_0^2 \dot{\gamma}_{0,T}^2 + (1+A^2v_0^2 \cos^2 \gamma_0)a_T^2}}{(1+A^2v_0^2 \cos^2 \gamma_0)}. \quad (19)$$

The sign of Eq. (19) is fundamentally determined by the direction of the IIP rate, with the downrange direction being assigned a positive sign. More precisely, if the direction of the IIP rate does not align (or is exactly opposite) with the direction of $\mathbf{i}_u^E$ in Eq.(12), the initially assigned sign should be reversed. Furthermore, the last component ($a_{T,3}$) can be computed from Eq. (14). Consequently, the acceleration command is expressed as

$$a_{T,2} = \dot{\gamma}_t v_0 - \left(\frac{v_0^2}{r_0} - \frac{\mu}{r_0^2} \right) \cos \gamma_0, \quad (20)$$

$$a_{T,1} = \frac{-ABv_0 \cos^2 \gamma_0 a_{T,2} \pm \sqrt{-(1+A^2v_0^2 \cos^2 \gamma_0 + B^2 \cos^2 \gamma_0)a_{T,2}^2 + (1+A^2v_0^2 \cos^2 \gamma_0)a_m^2}}{(1+A^2v_0^2 \cos^2 \gamma_0)}, \quad (21)$$

$$a_{T,3} = v_0 \cos \gamma_0 \left(Aa_{T,1} + B \frac{a_{T,2}}{v_0} \right). \quad (22)$$

Once the target flight-path-angle rate is determined, the acceleration command for the boost-back burn can be computed through Eqs. (20)-(22) without an iterative procedure. The flight-path-angle rate control scheme is activated when the flight-path-angle rate of the conventional IIP guidance algorithm falls below the predefined value. This study assumes that the optimal target flight-path-angle rate for the minimum fuel consumption is determined prior to the flight. Based on off-line trajectory optimization and various simulations with different target flight-path-angle rates, the value that minimizes propellant consumption is selected. Although a polynomial model could be used to represent the time-varying flight-path-angle rate, such variation over time may lead to unstable motion or inefficient performance in the presence of uncertainties in initial conditions, thrust level, or external disturbances. Future work could focus on real-time optimization of the target value to enhance mission flexibility and adaptability. Future work could focus on real-time optimization of the target value to enhance mission flexibility and adaptability.

3. Minimum Impulse Guidance (MIG)

Section 3 introduces the minimum impulse guidance (MIG) algorithm. Unlike the IIP guidance algorithm, whose acceleration command is directly derived from the IIP dynamics, this algorithm first computes the desired velocity that minimizes its increment while enabling the rocket to impact the target after the free flight. Subsequently, an acceleration command is generated using simple logic to follow the desired velocity. This algorithm can adopt two distinct approaches, contingent upon which of the two variables, 1) flight angle or 2) time of flight, is selected as the iterator variable to optimize the desired velocity. The choice of the iterator variable affects the efficiency of the algorithm, such as the propellant consumption and computational load.

3.1. Approach 1 – iterator variable: flight angle

The first approach selects the flight angle as an iterator variable. It first derives an expression for the desired velocity increment (Δv) as a function of the FPA in a 3-D frame for a given flight angle (target location in an inertial coordinate frame). Then, the algorithm determines an optimal FPA that minimizes the Δv by finding the zero of its gradient. Here, the TOF varies in accordance with the calculated velocity, and the target rotates during the TOF, meaning that the given flight angle does not guarantee the target location. Accordingly, we propose a targeting procedure (outer loop of this algorithm) to adjust the target location in the inertial frame until it matches the location calculated from the optimal FPA.

3.1.1. Optimal Desired Velocity Vector

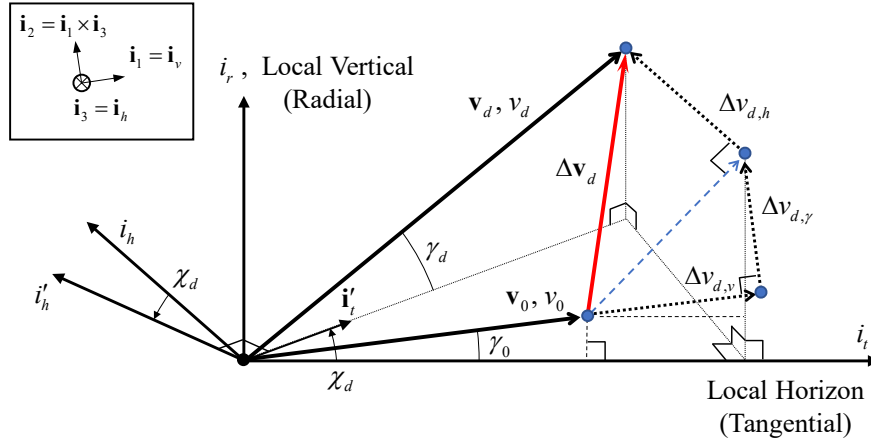


Fig. 4 Geometry of the desired Δv components

The *Hit Equation* describing the relationship between the desired flight angle (ϕ_d), flight speed (v_d), and FPA (γ_d) is expressed as [14]

$$\frac{r_0}{r_t} = \frac{1 - \cos \phi_d}{(r_0 v_d^2 / \mu) \cos^2 \gamma_d} + \frac{\cos(\phi_d + \gamma_d)}{\cos \gamma_d}. \quad (23)$$

Subscript d represents the desired values for the targeting, and r_t is the radius of the target position. Note that if the target is on the ground r_t is equal to r_p . The desired flight angle (ϕ_d) is expressed using the current and target vectors as

$$\phi_d = \cos^{-1} \left(\frac{\mathbf{r}_t \cdot \mathbf{r}_0}{|\mathbf{r}_t| |\mathbf{r}_0|} \right). \quad (24)$$

By rearranging Eq. (23), the desired velocity can be expressed as a function of γ_d as follows

$$v_d^2 = \frac{\mu r_t (\cos \phi_d - 1)}{r_0 \cos \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)}. \quad (25)$$

Eq. (25) implies that various possible combinations of the desired burn-out variables exist. The next step is to determine the particular combination that minimizes Δv_d . The desired velocity increment vector ($\Delta \mathbf{v}_d$) is defined as

$$\Delta \mathbf{v}_d = |\mathbf{v}_d - \mathbf{v}_0| = \sqrt{\Delta v_{d,v}^2 + \Delta v_{d,\gamma}^2 + \Delta v_{d,\chi}^2} \quad (26)$$

where $\Delta v_{d,v}$, $\Delta v_{d,\gamma}$, and $\Delta v_{d,\chi}$ are the components of $\Delta \mathbf{v}_d$ in $\mathbf{i}_1$, $\mathbf{i}_2$, and $\mathbf{i}_3$ directions, respectively, as shown in Fig. 4. We can simplify the problem by assuming the in-plane target position ($\mathbf{r}_t$). However, the target location can be off-plane due to errors, disturbances, and the Earth's rotation in practice. The variation is considerable, particularly in the RTLS scenario, requiring the consideration of the crossrange motion.

The objective function for the minimum Δv_d is:

$$\min_{\gamma_d, v_d} J = \Delta v_d^2 = (|\mathbf{v}_d - \mathbf{v}_0|)^2 = \Delta v_{d,v}^2 + \Delta v_{d,\gamma}^2 + \Delta v_{d,\chi}^2. \quad (27)$$

The geometry of Δv_d components in Fig. 4 provides the following relationships:

$$\Delta v_{d,v}^2 + \Delta v_{d,\gamma}^2 = (v_d \cos \gamma_d \cos \chi_d - v_0 \cos \gamma_0)^2 + (v_d \sin \gamma_d - v_0 \sin \gamma_0)^2, \quad (28)$$

$$\Delta v_{d,\chi} = v_d \cos \gamma_d \sin \chi_d, \quad (29)$$

where χ_d is the desired azimuth angle for the target location computed from the following equation:

$$\chi_d = \cos^{-1} (\mathbf{i}_h \cdot \mathbf{i}'_h). \quad (30)$$

In Eq. (30), $\mathbf{i}'_h$ is the angular momentum unit vector of the new plane formed by the current and target position vectors as

$$\mathbf{i}'_h = \frac{\mathbf{r}_0 \times \mathbf{r}_t}{|\mathbf{r}_0 \times \mathbf{r}_t|}. \quad (31)$$

Note that v_d and γ_d are the values defined in this new plane. Putting Eqs. (28)-(29) into Eq. (27) yields

$$\Delta v_d^2 = (v_d)^2 + (v_0)^2 - 2v_d v_0 (\cos \chi_d \cos \gamma_d \cos \gamma_0 + \sin \gamma_d \sin \gamma_0). \quad (32)$$

Consequently, applying Eqs. (25) and (30) into Eq. (32) gives

$$\begin{aligned} \Delta v_d^2 = & \frac{\mu r_t (\cos \phi_d - 1)}{r_0 \cos \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)} + v_0^2 \\ & - 2v_0 \left((\mathbf{i}_h \cdot \mathbf{i}'_h) \cos \gamma_d \cos \gamma_0 + \sin \gamma_d \sin \gamma_0 \right) \sqrt{\frac{\mu r_t (\cos \phi_d - 1)}{r_0 \cos \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)}}. \end{aligned} \quad (33)$$

Eq. (33) is an analytic expression for Δv_d as a function of γ_d only. The following step determines the optimal γ_d to minimize Δv_d expressed in Eq. (27). Taking the partial derivative of Eq. (33) to γ_d yields

$$\begin{aligned} \frac{\partial(\Delta v_d^2)}{\partial \gamma_d} = & -\frac{\mu r_t (\cos \phi_d - 1)(r_0 \sin 2\gamma_d - r_t \sin(\phi_d + 2\gamma_d))}{r_0 \cos^2 \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)^2} \\ & + \frac{\mu v_0 r_t ((\mathbf{i}_h \cdot \mathbf{i}'_h) \cos \gamma_d \cos \gamma_0 + \sin \gamma_d \sin \gamma_0)(\cos \phi_d - 1)(r_0 \sin 2\gamma_d - r_t \sin(\phi_d + 2\gamma_d))}{r_0 \cos^2 \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)^2} \sqrt{\frac{\mu r_t (\cos \phi_d - 1)}{r_0 \cos \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)}} \\ & + 2v_0 ((\mathbf{i}_h \cdot \mathbf{i}'_h) \sin \gamma_d \cos \gamma_0 - \cos \gamma_d \sin \gamma_0) \sqrt{\frac{\mu r_t (\cos \phi_d - 1)}{r_0 \cos \gamma_d (r_t \cos(\phi_d + \gamma_d) - r_0 \cos \gamma_d)}} \end{aligned} \quad (34)$$

From the optimality condition, the optimal γ_d should make the partial derivative in Eq. (34) zero. Due to the highly nonlinear form, it is impossible to derive an analytic expression for the optimal γ_d . Instead, the optimal desired FPA (γ_d^*) can be obtained by employing a root-finding technique (e.g., bisection method) with the upper and lower bounds expressed as [19]

$$\gamma_d^{\min} = \tan^{-1} \left[\frac{\cos \phi_d - r_0/r_t}{\sin \phi_d} \right], \quad \gamma_d^{\max} = \tan^{-1} \left[\frac{\sin \phi_d + \sqrt{2r_0/r_t(1 - \cos \phi_d)}}{1 - \cos \phi_d} \right]. \quad (35)$$

Note that the values at the bounds approaches infinity, causing numerical instability in their immediate vicinity. Narrowing the bounds by a small value resolves this issue. In [19], it is demonstrated that the desired velocity for a given flight angle is a unimodal (convex) function of the flight path angle within the bounds. Therefore, the partial derivative of the desired velocity in Eq. (34) has a unique root.

After finding the optimal velocity (v_d^*) from Eq. (25) with γ_d^* , we can calculate the optimal desired velocity vector in the ECI frame as

$$\mathbf{v}_d^* = (v_d^* \sin \gamma_d^*) \cdot \mathbf{i}_r + (v_d^* \cos \gamma_d^* \cos \chi_d) \cdot \mathbf{i}_t + (v_d^* \cos \gamma_d^* \sin \chi_d) \cdot \mathbf{i}_h. \quad (36)$$

3.1.2. Targeting Procedure

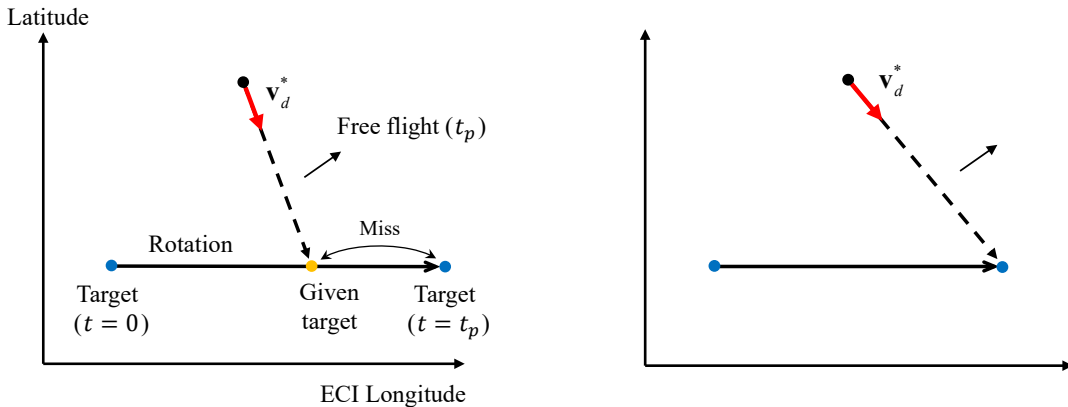


Fig. 5 Target location without targeting procedure (left) and targeting procedure (right)

The analysis in the previous subsection assumed that the target location in the ECI frame is given. However, the target rotates with the Earth, and thus, the target location in the ECI frame varies. The procedure described in the

previous subsection focuses on finding the optimal desired velocity without a constraint on TOF (t_p). In other words, the algorithm can guarantee the given target in the ECI frame, but the actual target moves (rotates) during t_p , which is determined based on $\mathbf{v}_d^*$. As a result, the mismatch occurs between the given and rotating targets, as depicted in Fig. 5-(a). To tackle this issue, this subsection introduces a targeting procedure.

The targeting procedure determines the target location in the inertial frame after the TOF (t_p), as shown in Fig. 5-(b). It starts with the rocket's current state, target position in the Earth-fixed frame (ECEF), and the initial guess of the flight angle (ϕ_d^{guess}). It is reasonable to use the flight angle defined in the ECEF frame as the initial guess of the flight angle. For the given flight angle, the longitude of the target in the inertial frame (Lon_t) is calculated as follows

$$\text{Lon}_t = \text{Lon}_0 + \cos^{-1} \left(\frac{\cos(\phi_d) - \sin(\text{Lat}_0) \sin(\text{Lat}_t)}{\cos(\text{Lat}_0) \cos(\text{Lat}_t)} \right) \quad (37)$$

where Lon_0 and Lat_0 are the longitude and latitude of the rocket in the ECI frame. The given target position can be computed as

$$\mathbf{r}_t = r_p [\cos(\text{Lat}_t) \cos(\text{Lon}_t) \quad \cos(\text{Lat}_t) \sin(\text{Lon}_t) \quad \sin(\text{Lat}_t)]^T. \quad (38)$$

Once the position is determined, the optimal desired velocity vector is calculated through the process described in subsection 3.1.1. Also, the TOF corresponding to the calculated optimal desired velocity vector is obtained from Eq. (9). As the method in subsection 3.1.1 does not impose a constraint on the TOF, the actual target position in the ECI frame differs from the given target. That is, the actual target rotates during the TOF, affecting its longitude as

$$\text{Lon}_t = \text{Lon}_t^E + \omega_E (t + t_p - t_{ref}), \quad (39)$$

where, Lon_t^E is the longitude of the target in the ECEF frame. From this new longitude of the target, the flight angle is updated from the following equation:

$$\cos \phi_d = \sin(\text{Lat}_0) \sin(\text{Lat}_t) + \cos(\text{Lat}_0) \cos(\text{Lat}_t) \cos(\text{Lon}_t - \text{Lon}_0) \quad (40)$$

If the difference between the previous and updated flight angles is smaller than the error threshold (ε), the procedure terminates. Then, the velocity obtained at the last iteration becomes the optimal desired velocity.

3.2. Approach 2 – iterator variable: time of flight

Approach 2 applies the Lambert problem to the boost-back guidance, leveraging its established framework for orbital mechanics. The unique aspect of this approach lies in applying this classical problem to compute the optimal time of flight (TOF) for minimum Δv during the boost-back maneuver, a use case that has not been extensively explored in previous works. This approach bridges traditional orbital mechanics and a real-time guidance algorithm, offering an efficient approach for achieving near-optimal performance under dynamic conditions.

Selecting the TOF as the iterator variable transforms the sub-problem of determining the desired velocity into the Lambert problem. Therefore, in contrast to approach 1, solving the sub-problem yields a feasible

and unique solution. It suggests that the velocity increment varies over the TOF, as shown in Fig. 6, necessitating an outer loop to select the optimal TOF that minimizes the desired velocity increment (Δv_d). This section presents the optimal (minimum Δv) boost-back guidance based on the solution of the Lambert routine and the optimal TOF search procedure.

This algorithm begins with computing the optimal TOF that minimizes Δv_d . Unlike approach 1, a targeting procedure is unnecessary in this approach since the solution of the Lambert problem assures targeting in the ECI frame for a given TOF. Various feasible solutions over the varying TOF exist (illustrated in the right figure in Fig. 6), and the corresponding velocity vector to each TOF is unique. As the right figure in Fig. 6 presents, the required Δv to change the current velocity to the desired velocity (obtained by solving the Lambert problem) is a convex function of a single variable, the TOF. Accordingly, we employ the golden-section search method to determine the optimal TOF corresponding to the minimum Δv . The right flowchart in Fig. 7 illustrates the procedure of the golden-section search applied to this algorithm. Thorough analysis defines an initial interval to encompass all potential variations of the optimal TOF during flight, and an interval of tens of seconds is considered a conservative choice.

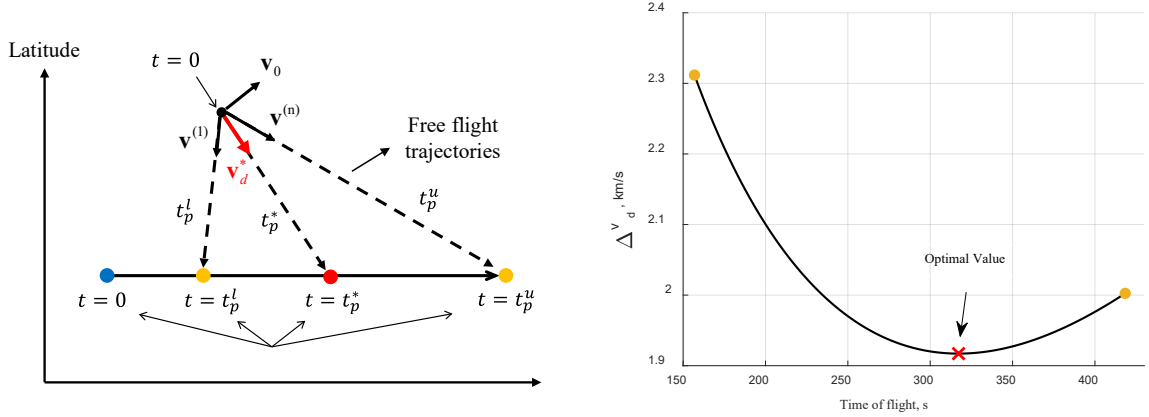


Fig. 6 Target change in ECI frame (left) and Δv versus time of flight (right)

This paper utilizes the straightforward steering algorithm to match the desired optimal velocity from the current velocity. The current and desired velocity vectors should coincide in the fastest time, and the simplest way to pursue this intent is to generate the acceleration command in the same direction as $\Delta \mathbf{v}_d^*$ expressed as

$$\mathbf{a}_T = \frac{\Delta \mathbf{v}_d^*}{\Delta v_d^*} a_m \quad (41)$$

To avoid chattering in the acceleration command near the target point, the burn is terminated when the error between the target point and the current IIP becomes smaller than tens of meters (set to 30 meters in the case study of this paper). This tolerance value can be adjusted depending on the mission and is corrected during the remaining two phases.

Fig. 7 compares the two approaches applied to the minimum impulse guidance. The difference between the two algorithms lies in the order of the routines where the minimum Δv is guaranteed. Approach 2 finds the optimal TOF and solves the Lambert problem with this value. Therefore, the optimality originates from the calculation of the optimal TOF. On the other hand, approach 1 directly calculates the optimal desired velocity ($\mathbf{v}_d^*$)

in every guidance cycle while determining the feasible TOF that ensures the target position.

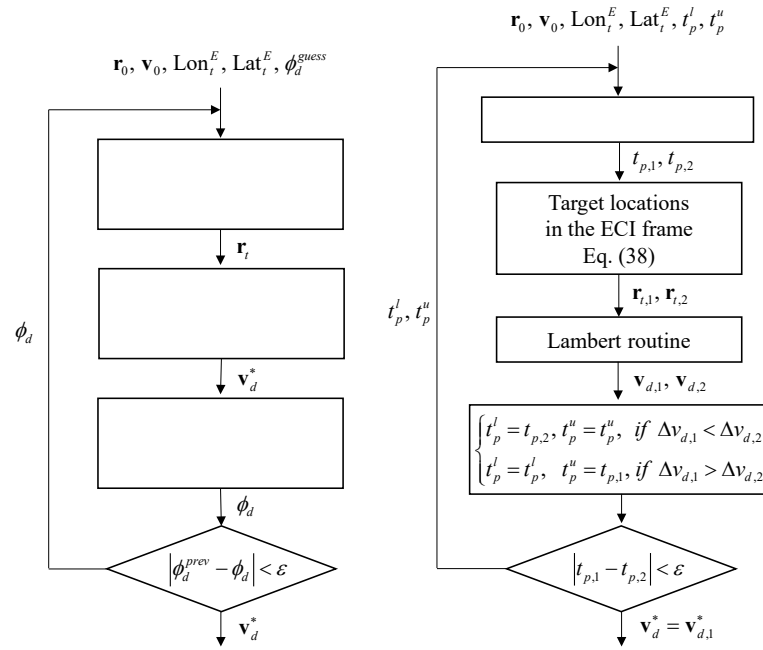


Fig. 7 Flowchart of the minimum impulse guidance with approach 1 (left) and approach 2 (right)

4. Case Study

We carried out a case study on the boost-back phase of a reusable stage with two landing scenarios to validate the three guidance algorithms. The two scenarios are the *downrange landing* (DRL) and the *return to launch site* (RTLS). We selected the Falcon 9 launch vehicle used for CRS 10 mission for the case study. The ECEF frame aligns with the ECI frame at the moment of lift-off, and the boost-back burn starts after 160 seconds from the lift-off. Therefore, the offset corresponding to this rotational interval occurs in the ECEF frame ($t_{ref} = -160$ s). Table 1 provides the specification and flight conditions of the reusable stage of Falcon 9 when the boost-back burn begins [20-21].

Table 1 Specification, initial condition, and target locations of the reusable stage

Rocket Specification	Value
Thrust, tonf	279.6
Specific impulse, s	311
Initial Condition	Value
Position ($\mathbf{r}_0$), [km, km, km]	[1051, -5541, 3160]
Velocity ($\mathbf{v}_0$), [km/s, km/s, km/s]	[1.475, 0.098, 1.395]
Mass (m_0), ton	79.2
Reference time (t_{ref}), s	-160
Target Site	Value
DRL ($\text{Lon}_t^E, \text{Lat}_t^E$), [deg, deg]	[-80.543, 28.486]
RTLS ($\text{Lon}_t^E, \text{Lat}_t^E$), [deg, deg]	[-78.450, 30.500]

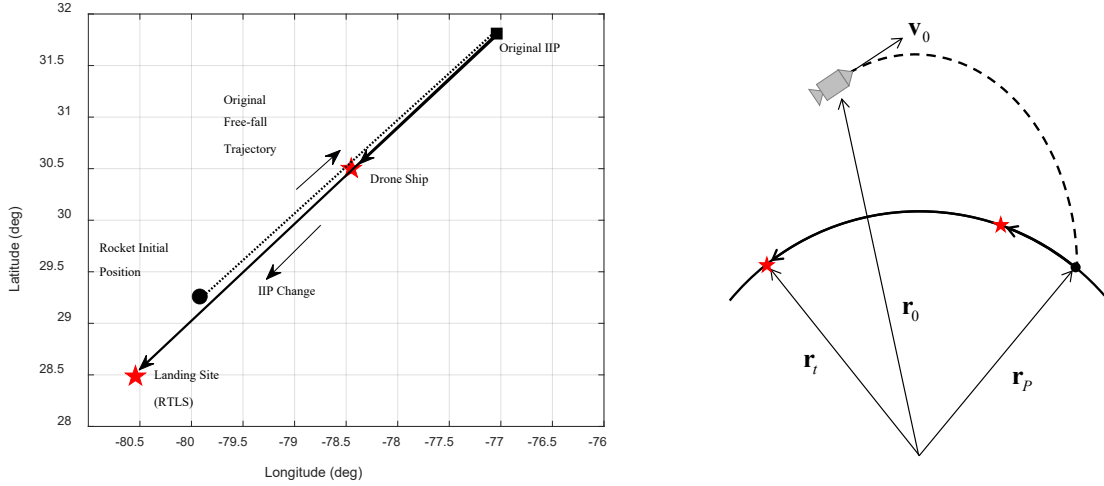


Fig. 8 Target locations in the ECEF frame

Fig. 8 presents the initial position/IIP of the rocket and target landing locations for the scenarios. The drone ship is located along the downrange direction, and the landing site for the RTLS scenario is in the opposite direction. Both scenarios require a pitch-over maneuver to move the IIP to the target locations. The proposed guidance algorithms are applied for the boost-back burn for each scenario, and the results are compared with the offline

trajectory optimization results [22]. The objective function of trajectory optimization is to minimize the burn time (final time), which is equivalent to the minimum propellant consumption. The target IIP condition is imposed as a constraint at the end of the boost-back burn. Therefore, the final IIP calculated from the position and velocity of the reusable stage at the final time should be equal to the target IIP. We employ Nelson and Zarchan's algorithm to solve the Lambert problem [19] and the IIP calculation routine described in Refs. [17-18]. Note that the flight-path-angle rate control scheme in the IIP guidance is activated when the angle rate is lower than the predesigned value. For conducting the case study of the IIP guidance algorithm considering the FPA in the RTLS scenario, we select 0.6 deg/s as the desired FPA rate through various guidance simulations. This value could alter depending on the scenario. In the DRL scenario, as the FPA does not decrease, the FPA rate control scheme is not activated.

4.1. Scenarios – Return to Launch Site (RTLS) and Down Range Landing (DRL)

Table 2 compares the results (burn time) obtained from the proposed guidance algorithms, the existing IIP guidance algorithm, and offline trajectory optimization. The proposed algorithms show near-optimal performances in both scenarios compared to the optimal results: the optimality gaps for the DRL scenario are insignificant (0.2 ~ 0.9 %). The maximum optimality gap occurs in the RTLS scenario performed by the AIIP guidance, but the value is only 3.9 %. By effectively addressing the issue of downward trajectories, the proposed method achieves significantly improved performance over the existing IIP guidance in the RTLS scenario. In cases where a downward trajectory does not occur (e.g., DRL), the flight-path-angle rate control routine is not activated, resulting in identical performance to the IIP guidance.

Table 2. Case study results: proposed algorithms versus optimal results

Descent Scenario	Off-line trajectory optimization	Burn time, s (difference, %)			
		Original law		Proposed laws	
		IIP guidance [9]	IIP guidance w/ FPA	Minimum Impulse Guidance	
				Approach1	Approach 2
DRL	17.59	17.70 (0.6)	17.70 (0.6)	17.76 (0.9)	17.62 (0.2)
RTLS	43.29	54.39 (25.6)	44.96 (3.9)	43.75 (1.1)	43.35 (0.1)

Figs. 9-12 present the case study results for the DRL scenario. As seen in Fig. 9, all guidance algorithms successfully change the IIP to the target location (drone ship) with IIP trajectories similar to the optimal solution. Fig. 10 plots the altitude versus downrange during the boost-back burn, and Fig. 11 plots the FPA and velocity profiles. The trajectory obtained by the minimum impulse guidance with approach 2 is the closest to the optimal trajectory. Fig. 12 plots the acceleration command profiles. Despite the similarity in the ground trajectory, we can observe that the states and acceleration profiles for each algorithm are notably different.

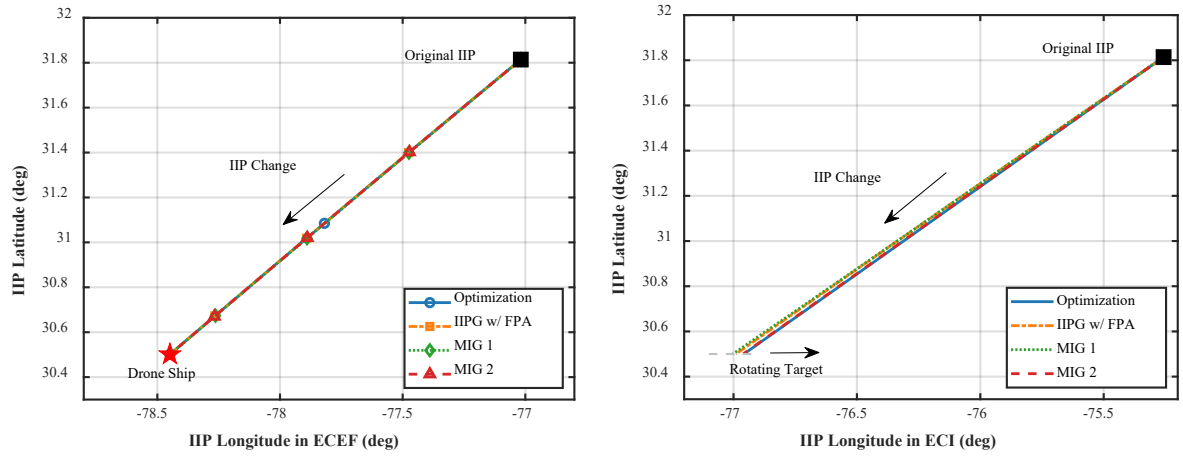


Fig. 9 IIP ground trajectories in ECEF (left) and ECI (right) frames (DRL scenario)

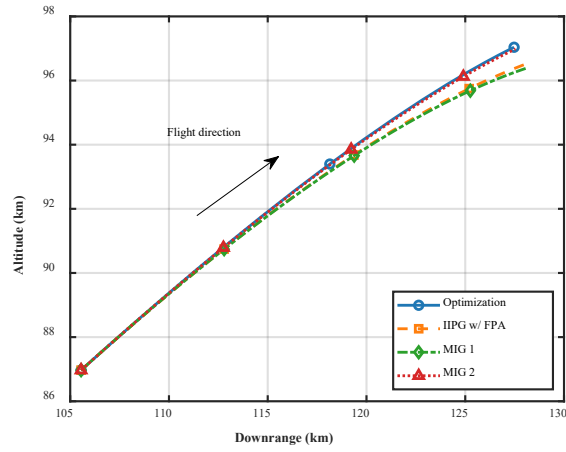


Fig. 10 Altitude versus downrange of the reusable stage (DRL scenario)

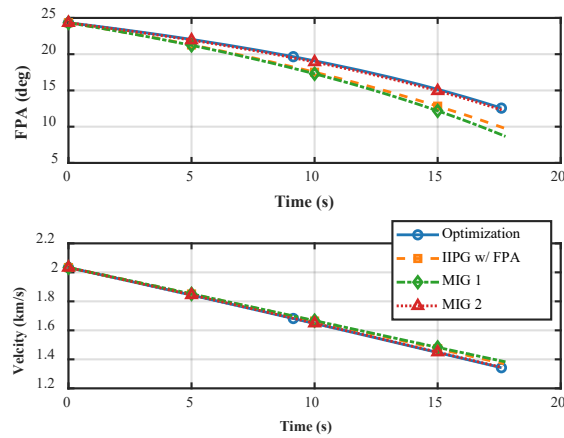


Fig. 11 Flight path angle and velocity profiles of the reusable stage (DRL scenario)

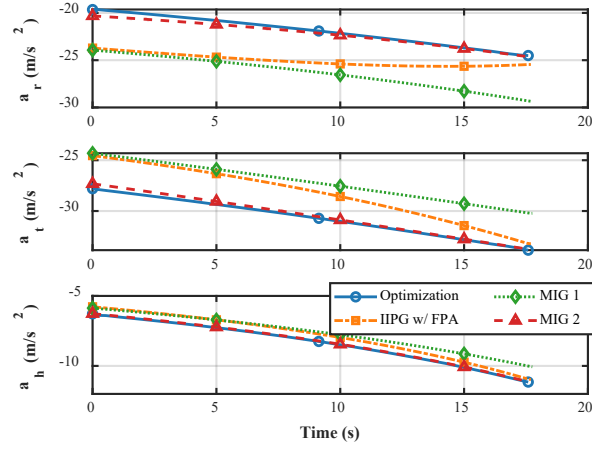


Fig. 12 Acceleration command profiles (DRL scenario)

Figs. 13-16 present the case study results for the RTLS scenario. Fig. 13 indicates that all algorithms can be applied to the boost-back phase in the RTLS scenario in that the IIP moves to the target position. The difference in the flight causes the difference in trajectories expressed in the ECI frame shown in Fig. 13. Figs. 14-15 show the guided boost-back trajectories, FPA, and velocity profiles obtained from the proposed algorithms / offline optimization. Fig. 16 presents the acceleration command profiles. We can observe that the variations in the trajectories/states for the RTLS case are larger than those for the DRL case, explaining a relatively large optimality gap (presented in the last row of Table 2).

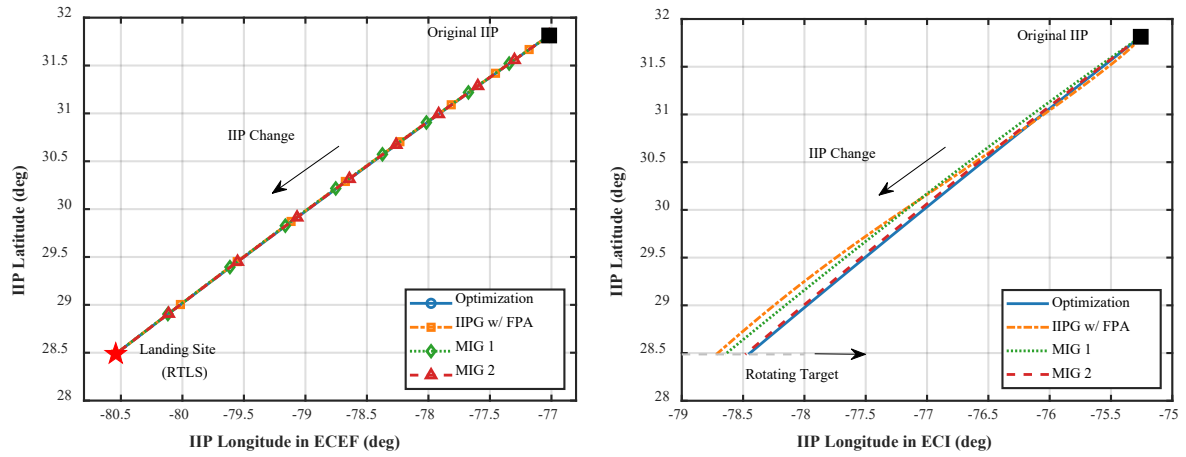


Fig. 13 IIP trajectories in ECEF (left) and ECI (right) frames (RTLS scenario)

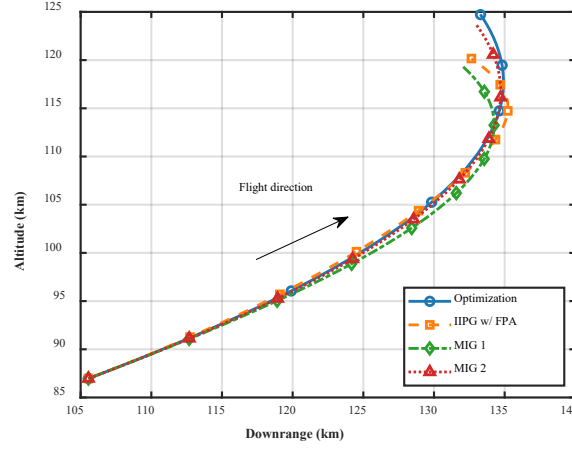


Fig. 14 Altitude versus downrange of the reusable stage (RTLS scenario)

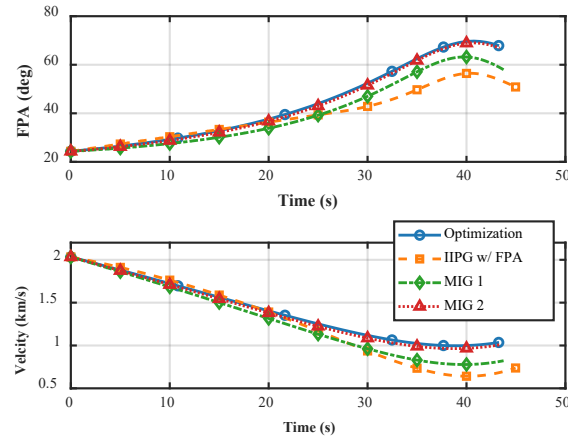


Fig. 15 Flight path angle and velocity profiles of the reusable stage (RTLS scenario)

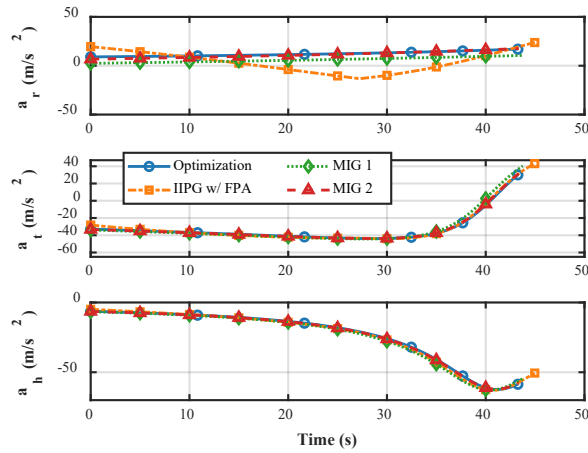


Fig. 16 Acceleration command profiles (RTLS scenario)

4.2. Monte Carlo Simulation

We performed a Monte Carlo simulation with 500 runs for the RTLS scenario of each algorithm to evaluate the robustness of the proposed algorithms. The simulation considers uncertainties in thrust and specific

impulse, each modeled as a normal distribution with a 3σ deviation equivalent to 5% of the nominal value as represented in Table 3. Additionally, initial state deviations with 3σ of 5 km in position and 50 m/s in velocity are applied to account for realistic state deviations [13, 23].

Table 3. Uncertainties in thrust, specific impulse, and initial states

	Thrust	I_{sp}	Position	Velocity
3σ Range	$\pm 5\%$	$\pm 5\%$	$\pm 5\text{km}$	$\pm 50\text{m/s}$

Table 4 summarizes the Monte Carlo simulation results, including the mean and standard deviation of targeting error. The results show that the proposed guidance algorithms successfully move the IIP to the landing site. These results indicate that the proposed guidance algorithms are robust to the initial state deviations and the engine's performance uncertainty, demonstrating their adaptability for practical applicability.

Table 4. Miss Distance Results from 500 Monte Carlo Runs

Miss distance	IIP guidance w/ FPA	Minimum Impulse Guidance	
		Approach 1	Approach 2
Mean (m)	4.4	4.3	4.7
Standard deviation (m)	2.5	2.5	2.9

Table 5 compares the computational costs of the proposed algorithms. The absolute computational time varies by CPU performance, so showing that the algorithms are computationally efficient with high-end CPUs may be unfair. Instead, we present the computational cost (mean and maximum time) relative to the IIP guidance considering FPA, which has no iterative procedure and is therefore obviously computationally efficient. Accordingly, we measured the computational time elapsed to run each guidance loop and calculated the mean and max values on a desktop computer with a 16-core processor (3.40 GHz) and 64 GB of RAM. The mean and max values of the computational time for the minimum impulse guidance algorithms are larger than the IIP guidance considering FPA by 73% and 229%, respectively. The maximum computation time is observed in the first loop, where the initial guesses are rough. Once the algorithm starts using the previous values as the initial guesses at each cycle, the solution converges much faster in the minimum impulse guidance. In contrast, approach 2 shows a slower result in both indices. However, given that the IIP guidance algorithm is iteration-free and computationally efficient, a slowdown of 3 to 4 times may not be a substantial issue. This is supported by a recent study [24] which emphasized numerical guidance algorithms, inherently reliant on iterative processes, have become mainstream as onboard computational efficiency has drastically improved. The results show that the proposed algorithms significantly outperform the off-line trajectory optimization, which requires several seconds to compute, in terms of computation time, while achieving near-optimal propellant consumption.

Table 5 Relative computational cost of proposed boost-back guidance algorithms

	IIP guidance w/ FPA	Minimum Impulse Guidance	
		Approach 1	Approach 2
Mean	1 (1.7 μs)	1.73	3.29
Max	1 (32.6 μs)	2.04	3.10

Table 6 summarizes the characteristics of the proposed algorithms. The core strength of the IIP-based guidance algorithm lies in its iteration-free nature and the capability to account for the FPA or velocity constraints at a specific altitude (e.g., initial velocity constraint of the re-entry phase) [25]. Leveraging these strengths, future research could explore improving the algorithm by considering favorable initial conditions of the subsequent phases. However, this algorithm requires the reference value ($\dot{\gamma}_t$) in the RTLS scenario, which may deteriorate operational flexibility and adaptability. Its optimality gap is relatively large compared to the minimum impulse guidance, while the value (3.9 %) is acceptable.

The minimum impulse guidance algorithm does not require any reference value. This algorithm generates the acceleration command solely through the current position and velocity feedback, providing operational flexibility against disturbances. Moreover, the case study results show the remarkable optimality of these algorithms. On the other hand, the disadvantages can be found in the computational time and the necessity for iterations. Various tests for the proposed algorithms, including worst-case performance assessments, should be conducted to warrant their onboard applicability.

Table 6. Characteristics comparison of the proposed guidance algorithms

Algorithm	Advantage	Drawback
IIP guidance w/ FPA	Non-iterative Capable of considering constraints	Reference value ($\dot{\gamma}_t$) needed (RTLS) Relatively large optimality gap
Minimum impulse guidance	No reference values needed High flexibility and adaptability Remarkable optimality	Iteration inherent Hard to impose constraints

5. Conclusion

This paper proposes two types of feedback guidance algorithms for the boost-back burn of a reusable launch vehicle. The first algorithm, FPA-constrained IIP guidance, augments the conventional IIP guidance algorithm to extend its adaptability by addressing the drawbacks. The second algorithm, minimum impulse guidance, yields the optimal desired velocity based on the analytic expressions or the Lambert problem, minimizing the velocity increment. Case studies with a realistic reusable launch vehicle show that all algorithms show acceptable performance in terms of the computational time and optimality gap. Moreover, we investigate the characteristics of the proposed algorithms. Given the results, we can conclude that the IIP guidance is most suitable in terms of both computational time and propellant consumption for DRL scenario, while the RTLS scenario may need a

trade-off study between the two indices as both algorithms exhibit contrasting characteristics. An in-depth analysis of their applicability for onboard implementation, consideration of the favorable initial conditions of the remaining phases (re-entry and terminal landing phases), and virtual targeting strategy considering aerodynamic forces can be a promising subject for future study

Acknowledgment

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Appendix: Derivation of Instantaneous Impact Point Rate

The IIP changes when the external forces, such as thrust, act on the rocket while the IIP of a rocket, which is only affected by gravity, remains unchanged. The IIP rate vector ($\dot{\mathbf{r}}_p$) induced from the thrust can be decomposed into two terms: downrange direction component ($\dot{\mathbf{r}}_{p,D}$, caused by in-plane motion) and crossrange direction component ($\dot{\mathbf{r}}_{p,C}$, caused by plane change motion) expressed as

$$\dot{\mathbf{r}}_p = \dot{\mathbf{r}}_{p,D} + \dot{\mathbf{r}}_{p,C}, \quad (42)$$

$$\dot{\mathbf{r}}_{p,D} = r_p \left(\frac{\partial \phi}{\partial v_0} \dot{v}_{0,T} + \frac{\partial \phi}{\partial \gamma_0} \dot{\gamma}_{0,T} \right) \cdot \mathbf{i}_D, \quad (43)$$

$$\dot{\mathbf{r}}_{p,C} = r_p (\sin \phi \cdot \dot{\chi}_{0,a}) \cdot \mathbf{i}_C, \quad (44)$$

where,

$$\mathbf{i}_D = -\frac{\sin(\phi + \gamma_0)}{\cos \gamma_0} \mathbf{i}_{r_0} + \frac{\cos \phi}{\cos \gamma_0} \mathbf{i}_{v_0}, \quad \mathbf{i}_C = \mathbf{i}_h = \mathbf{i}_3. \quad (45)$$

Then, the rate of the IIP unit vector can be obtained by combining Eqs. (42)-(44) as

$$\begin{aligned} \frac{d}{dt} \mathbf{i}_p &= \frac{\dot{\mathbf{r}}_p}{r_p} = \left(\frac{\partial \phi}{\partial v_0} \cdot \mathbf{i}_D \right) \dot{v}_{0,T} + \left(\frac{\partial \phi}{\partial \gamma_0} \cdot \mathbf{i}_D \right) \dot{\gamma}_{0,T} + (\sin \phi \cdot \mathbf{i}_C) \dot{\chi}_{0,a} \\ &= \mathbf{d}_v \dot{v}_{0,T} + \mathbf{d}_\gamma \dot{\gamma}_{0,T} + \mathbf{d}_\chi \dot{\chi}_{0,T}, \end{aligned} \quad (46)$$

where,

$$\frac{\partial \phi}{\partial v_0} = \frac{2\mu(1 - \cos \phi)}{v_0 \mu \sin \phi - \frac{1}{2} r_0 v_0^3 [\sin(2\gamma_0 + \phi) + \sin \phi]}, \quad (47)$$

$$\frac{\partial \phi}{\partial \gamma_0} = \frac{\sin(2\gamma_0 + \phi) - (r_0 / r_p) \cdot \sin 2\gamma_0}{\frac{\mu \sin \phi}{r_0 v_0^2} - \frac{1}{2} [\sin(2\gamma_0 + \phi) + \sin \phi]}. \quad (48)$$

Three sensitivity vectors In Eq. (46) ($\mathbf{d}_v$, $\mathbf{d}_\gamma$ and $\mathbf{d}_\chi$) represent the influence of the velocity rate

components on the IIP rate vector. The Earth rotation effect should be considered in calculating the IIP rate to target the desired position as

$$\frac{d}{dt}\mathbf{i}_p^E = \mathbf{C}_I^E \left(\frac{d}{dt}\mathbf{i}_p \right) - \dot{t}_{p,a} (\boldsymbol{\Omega} \times \mathbf{i}_p^E), \quad (49)$$

where, $\boldsymbol{\Omega}(\equiv [0, 0, \omega_E]^T)$ is the rotation vector of the Earth with the angular velocity of ω_E , and $\mathbf{C}_I^E$ is the transformation matrix from the ECI frame to the ECEF frame given as

$$\mathbf{C}_I^E = \begin{bmatrix} \cos(\omega_E \Delta t_R) & \sin(\omega_E \Delta t_R) & 0 \\ -\sin(\omega_E \Delta t_R) & \cos(\omega_E \Delta t_R) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (50)$$

In Eq. (50), Δt_R is the time difference between the two frames when the rocket arrives at the IIP expressed as

$$\Delta t_R = t - t_{ref} + t_p, \quad (51)$$

where t is the current time and t_{ref} is the time when the two frames (ECI and ECEF) align. In Eq. (49), superscript E represents the values in the ECEF frame and $\dot{t}_{p,a}$ is the time derivative of the TOF caused by the external acceleration defined as

$$\dot{t}_{p,T} = D_{\dot{v}}^{i_p} \dot{v}_{0,T} + D_{\dot{\gamma}}^{i_p} \dot{\gamma}_{0,T}, \quad (52)$$

where,

$$D_{\dot{v}}^{i_p} = D_{\dot{a}}^{i_p} \frac{2a^2 v_0}{\mu} + D_{\dot{e}}^{i_p} \left(\frac{p v_0}{\mu e} - \frac{r_0^2 v_0 \cos^2 \gamma_0}{\mu a e} \right), \quad D_{\dot{\gamma}}^{i_p} = D_{\dot{e}}^{i_p} \frac{r_0^2 v_0^2 \cos \gamma_0 \sin \gamma_0}{\mu a e}, \quad (53)$$

$$D_{\dot{a}}^{i_p} = \frac{\partial \dot{t}_p}{\partial \dot{a}} = \frac{3t_p}{2a} - \frac{1}{a^2 e n} \left[\frac{r_e (1 - e \cos E_p)}{\sin E_p} - \frac{r_0 (1 - e \cos E_0)}{\sin E_0} \right], \quad (54)$$

$$D_{\dot{e}}^{i_p} = \frac{\partial \dot{t}_p}{\partial \dot{e}} = \frac{1}{n} \left[\left(\frac{\cos E_p (1 - e \cos E_p)}{e \sin E_p} - \frac{\cos E_0 (1 - e \cos E_0)}{e \sin E_0} \right) - (\sin E_p - \sin E_0) \right]. \quad (55)$$

Details on the orbital parameters (a , e , p , n , E_0 , and E_p) in Eqs. (53)-(55) are available in Refs. [17-18]. Putting Eqs. (46) and (52) into Eq. (49) yields

$$\frac{d}{dt}\mathbf{i}_p^E = \mathbf{d}_{\dot{v}}^E \dot{v}_{0,T} + \mathbf{d}_{\dot{\gamma}}^E \dot{\gamma}_{0,T} + \mathbf{d}_{\dot{\chi}}^E \dot{\chi}_{0,T}, \quad (56)$$

where,

$$\mathbf{d}_{\dot{v}}^E = \omega_E D_{\dot{v}}^{i_p} \begin{bmatrix} \dot{i}_{py}^E \\ -\dot{i}_{px}^E \\ 0 \end{bmatrix} + \mathbf{C}_I^E \mathbf{d}_{\dot{v}}, \quad \mathbf{d}_{\dot{\gamma}}^E = \omega_E D_{\dot{\gamma}}^{i_p} \begin{bmatrix} \dot{i}_{py}^E \\ -\dot{i}_{px}^E \\ 0 \end{bmatrix} + \mathbf{C}_I^E \mathbf{d}_{\dot{\gamma}}, \quad \mathbf{d}_{\dot{\chi}}^E = \mathbf{C}_I^E \mathbf{d}_{\dot{\chi}}. \quad (57)$$

Three sensitivity vectors ($\mathbf{d}_{\dot{v}_a}^E$, $\mathbf{d}_{\dot{\gamma}_a}^E$ and $\mathbf{d}_{\dot{\chi}_a}^E$) in Eq. (56) specify the influence of the velocity rate components on the IIP rate defined in the ECEF frame. Eq. (56) can be expressed with the thrust acceleration components by using Eq. (3) as follows

$$\frac{d}{dt} \mathbf{i}_p^E = \mathbf{d}_{a_{T,1}}^E a_{T,1} + \mathbf{d}_{a_{T,2}}^E a_{T,2} + \mathbf{d}_{a_{T,3}}^E a_{T,3} \quad (58)$$

where

$$\mathbf{d}_{a_{T,1}}^E = \mathbf{d}_{\dot{v}}^E, \quad \mathbf{d}_{a_{T,2}}^E = \frac{\mathbf{d}_{\dot{\gamma}}^E}{v_0}, \quad \mathbf{d}_{a_{T,3}}^E = \frac{\mathbf{d}_{\dot{\chi}}^E}{v_0 \cos \gamma_0}. \quad (59)$$

More details on the procedure deriving the IIP dynamics can be found in Refs. [17-18].

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